Module 5A – The Earth's Atmosphere



Figure 1

We live on the Earth, the most remarkable place in the Solar System, and (so far) the only "rock" in the entire Universe that is known to support life.

In this module, we will look at the Earth's **atmosphere**, not only its present form, but also the *evolution* of it to its present form. In doing so, we will look at some characteristics of Earth's ground and space weathers.

We define the current **atmosphere** as

> the gaseous envelope that encapsulates the Earth.

Excluding water vapor, Earth's current atmosphere consists primarily of the following **four gases**:

- 78% Nitrogen (N_2),
- 21% Oxygen (O_2),
- .93% Argon (Ar), and
- .04% Carbon Dioxide (CO_2)

Notice how little of our current atmosphere is actually composed of carbon dioxide.

The current atmosphere is divided into *four layers* that are separated by relatively thin boundaries, called pauses, where the temperature trend reverses.

Layer 1: The Troposphere

The **troposphere** is the first layer of the Earth's atmosphere, closest to the surface of the Earth.

How high the troposphere ascends depends on Earth latitude, with the top of the troposphere being

- 17 km (10.6 mi, or 56,000 ft) in the tropics,
- 11 km (6.8 mi, or 36,000 ft) at mid-latitudes, and
- 8 km (5 mi, or 26,000 ft) at the north and south poles

At mid-latitudes, the troposphere contains about 78% of Earth's air by mass (about 90% in the tropics), and 99% of the air's water vapor.

Air temperatures decrease with increasing height within the troposphere. At sea level, the average air temperature worldwide is 15° C (59° F). At the top of the troposphere,

called the **tropopause**, the average temperature of the air dips to -56° C (-69° F) at mid-latitudes, and to about -78° C (-109° F) in the tropics. Above the tropopause, the air *starts to warm again*.

The pressure of air at sea level is 101,325 Pascals (Pa), or 14.7 pounds per square inch (psi).

Virtually *all of Earth's weather* (that is not space based) occurs in the troposphere.

At mid-latitudes, commercial jets travel in the upper troposphere and lowest stratosphere, typically at altitudes of 30,000 ft to 41,000 ft, with business jets going as high as 45,000 ft or more. One exception: the now defunct Concorde supersonic commercial jet, which regularly flew at 60,000 ft, well above the tropopause.

In the *northern hemisphere*, just below the tropopause, there are two west-to-east flows of high-speed winds called the **jet streams**.

One stream, called the sub-tropical jet stream, is approximately located at 30° N latitude. The other stream, called the polar jet stream, is approximately located at 50° to 60° N latitude.

When possible and convenient, commercial jet aircraft moving west to east like to "ride" one of these jet streams as a tail wind to lessen the plane's travel time (and save jet fuel). Normal jet stream winds move at speeds between 129 kph and 225 kph (80 mph and 140 mph).



Figure 2

Hot air balloons have flown as high as 69,000 ft (a 2005 record), but seldom have they ventured into either of the jet streams.
As a point of comparison, the highest-flying bird, Rüppell's Griffon Vulture, flies at the astounding height of 37,000 ft.

Because there is a relatively sudden change in temperature at the tropopause, cumulonimbus clouds (which are violent rain storm clouds) usually terminate, in a flat-topped fashion (i.e. anvil out) near the tropopause, at about 36,000 to 45,000 ft at mid-latitudes, and up to about 56,000 ft in the tropics.

Layer 2: The Stratosphere

The **stratosphere** is the second layer of the Earth's atmosphere, just above the tropopause.

At mid-latitudes, the stratosphere extends from the tropopause up to a height of about 50 km (31 mi, or 164,000 ft). This height is known as the **stratopause**.

The stratosphere and troposphere together contain 99.9% of the Earth's air by mass.

Counterintuitively, **air temperatures rise (**or stay steady) with increasing altitude in the stratosphere, due to interactions between ozone (O_3), oxygen (O_2), and the Sun's ultraviolet (UV) radiation, with temperatures at the stratopause reaching a comparatively balmy high of about -2° C (28° F).

In winter, in the polar region of the stratosphere, water vapor freezes immediately as extremely small ice crystals, forming nacreous clouds that can be seen as they reflect light from the Sun during civil twilight.

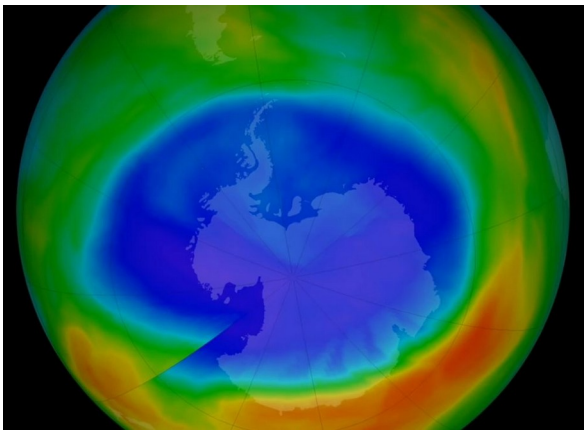


Figure 3

Ozone accumulates in a layer in the stratosphere, starting at about 15 km and thinning out toward the stratopause (50 km), with a peak abundance at about 25 km altitude.

Ozone and oxygen attenuate the more destructive wavelengths of the Sun's ultraviolet (UV) radiation, radiation that would cause DNA damage to life (if it were to get to Earth ground).

Concern over depletion of the ozone layer, primarily a hole that developed at the south pole due to humankind's release of chlorofluorocarbons (CFCs), resulted in a ban of these compounds. Fortunately, the ozone hole has slowly started to "heal" (or close up).

Ozone can also occur in the troposphere, where it is considered a pollutant that can seriously impair human lung function.

Lightning related events called blue jets occur in the stratosphere, directly above thunderstorms happening in the upper troposphere.

At the stratopause, the density of air is one thousandth air's density at sea level, *making oxygen-intake based jet flight impossible*.

High-altitude reconnaissance jets, like the U-2, fly in the *lower* stratosphere, at altitudes of 60,000 ft to 70,000 ft. An even higher flier was the US spy plane, the SR-71 (or Blackbird), now decommissioned, which flew up to 85,000 ft.

For comparison, Alan Eustace sky dived from a helium filled balloon, from a record height of just under 136,000 ft. You can watch his flight and Ted Talk here.

Layer 3: The Mesosphere

The **mesosphere** is the third layer of the Earth's atmosphere, just above the stratopause.

At mid-latitudes, the mesosphere extends from the stratopause up to a height of about 85 km (53 mi, or 279,000 ft). This height is known as the **mesopause**.

Air temperatures decrease with increasing altitude, reaching the blisteringly coldest temperature on or near Earth of -137° C (-215° F), reached above the poles in summer. At mid-latitudes, the mesopause averages about -100° C (-148° F).

Water vapor produced by the decomposition of methane in the mesosphere freezes immediately as extremely small ice crystals, forming noctilucent clouds that can be seen as they reflect light from the Sun during astronomical twilight.

The **ionosphere** is a region of the Earth's atmosphere, *starting* in the mesosphere (at about 60 km height) and extending to the top of the thermosphere (about 600 km). In the ionosphere, solar radiation creates a layer of electrons and positive ions (a plasma) that can reflect some (but not all) radio waves originating on Earth.

This reflection, known as skywave or skip, reflects radio waves up to a frequency of 30 million Hertz (30 MHz), and is used by "shortwave" radio enthusiasts and "ham radio" operators to communicate with each other around the world.



Figure 4

On moonless nights, when the Sun has long set, a faint greenish glow can be seen low along the horizon, called airglow, caused by the recombination of atoms and molecules that were ionized or split apart by sunlight during the day. (Image is from the Space Station.)



Figure 5

Lightning related events called red sprites occur in the mesosphere, and are triggered by powerful cloud-to-ground lightning strikes in the thunderstorms below.

Near the top of the mesosphere, at about 90 km, is a layer of sodium atoms that can be stimulated by finely tuned laser beams so as to glow like a star. These glowing sodium atoms are called laser guide stars, and are used by Earth based telescopes to adapt reflecting parts of the scopes so as to create images that are clear of the distortions caused by turbulence in the Earth's atmosphere directly above the scopes.

Lastly, **meteoroids** (small space rocks that enter Earth's atmosphere) typically burn up in the mesosphere as meteors, or shooting stars.

Layer 4: The Thermosphere

The **thermosphere** is the fourth and last true layer of the Earth's atmosphere. It starts at the mesopause and extends out a variable but typically maximum distance of 600 km (or 373 mi) called the **thermopause**.

Like the stratosphere, **air temperatures rise** with increasing altitude in the lower thermosphere, with maximum temperatures depending on sunspot activity.

For high sunspot activity, and when the Sun is shining, the maximum temperature in the Thermosphere is approximately 2000° C (3632° F).

For low sunspot activity, and when the Sun is shining, the maximum temperature in the thermosphere is approximately 500° C (932° F).

The **ionosphere** is densest in the thermosphere, peaking at about 300 km (186 mi), and ionization occurs in all of the thermosphere, due mostly to collisions between the few air molecules in the thermosphere, and very shortwave ultraviolet radiation and x-ray radiation from the Sun.



Figure 6

The Earth's protective magnetic field (the magnetosphere) electromagnetically traps charged particles coming from the Sun (the solar wind), with those particles spiraling downward towards the Earth near the north and south poles. Those particles interact with the air in the thermosphere, at altitudes ranging from about 100 km (62 mi) to 400 km (250 mi), causing the familiar northern (aurora borealis) and southern (aurora australis) lights. The *green and red colors* of the aurorae (plural of aurora) come from excited oxygen atoms, whereas the *blue and violet colors* of the aurorae come from excited nitrogen molecules.

At a height of 100 km (62 mi), the Kármán line, the atmosphere is so thin that it is considered (at least by the Fédération Aéronautique Internationale, which certifies aviation and spaceflight records) the beginning of outer space.

Further out, at 160 km (99 mi), where the anacoustic zone begins, the air is so thin that regular sound transmission is not possible.

The International Space Station (ISS) orbits Earth within the thermosphere, at a comfortable distance of about 408 km (254 mi).

Sometimes an extra layer is added to the traditional four layers of the Earth's atmosphere, the exosphere, but we will not discuss this so-called "extra layer" here.

Earth Based Weather in the Troposphere

As mentioned earlier, all Earth based weather occurs in the troposphere. We can think of **Earth based weather** as

> thermodynamic events occurring in the atmosphere.

By thermodynamic events, we mean events driven by the movement of heat, due to a difference in temperatures. By atmosphere, we mean the troposphere.

There are **six main ingredients** of Earth based weather:

- the atmosphere
- water (H_2O, in all three phases)
- the Sun
- the rotation of the Earth
- volcanic activity
- biological activity

The Atmosphere

Meteorology is

> the study of weather.

The word origin of meteorology is "the study of the atmosphere." It seems clear that without an atmosphere, a planet or moon cannot have weather. Our Moon is a classic example of this obvious truth.

Water

Water, dihydrogen oxide (H_2O) in all of it three common phases (ice, liquid water, and water vapor), plays perhaps the most crucial role in weather events.

Precipitation (in weather)

> is water falling from the sky.

Water falls in liquid form as rain. It can fall as fully formed ice crystal structures known as snow, or it can fall in a combination of rain and snow called sleet.

In violent storms involving cumulonimbus clouds, falling rain can freeze in pellet form and then be pushed up and down by drafts of air for relatively long times, forming what

is called hailstones. Hailstones vary in size, from 5 mm (0.2 in) to as large as 20 cm (8 in).



Figure 7

Lightning and thunder commonly occur in cumulonimbus rain storms.

Lightning is

an electrical discharge between air molecules ionized by collisions.

Lightning can be cloud to cloud, cloud to ground, or ground to cloud.

Lightning can cause the formation of nitrates in rain water which subsequently fertilize vegetation below. This is why grass is noticeably greener after a storm with lightning.

Thunder is

a shockwave sound that accompanies lightning.

Since “sound in air” travels much more slowly than “light in air”, you can determine *how far away a lightning strike occurred* by counting the seconds between a lightning flash and the subsequent sound of thunder. Each second of sound delay corresponds to a distance of about 1/5th mile.

All precipitation occurs in clouds, in the form of very tiny water droplets (or in some cases, ice crystals) that have formed around some nucleating dust or bacteria. The water droplets form because the relative humidity in the cloud has reached 100%. The temperature at which a parcel of air reaches 100% relative humidity (and a cloud forms) is called the dew point temperature.

Under the influence of wind (or another factor, like a “cold front” or the side of a mountain), the water droplets in clouds coalesce (are pushed together) to the point of being too heavy to stay aloft, and precipitation commences.

Clouds appear in the **homosphere**, which is

the troposphere, the stratosphere and the mesosphere.

Altogether, there are *10 basic cloud types (genera)*, including fog, which is just a cloud at ground level.

How then does liquid water get into the air to form clouds and result in precipitation? The answer is that the liquid water literally “jumps into the air,” molecule by molecule, in a process known as evaporation.

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Evaporation of liquid water is

> the persistent transfer of fast-moving water molecules from liquid water to the air directly above it.

Liquid water molecules naturally have a distribution of speeds, and fast-moving water molecules at the “water to air” boundary of a puddle of water have enough speed to literally “jump out of the water” into the air, becoming water vapor molecules.

In doing so, the water molecules take kinetic energy with them. We say that

> evaporation is a cooling process.

You might think that evaporation would necessarily cease after “all the fast molecules” had jumped out of the water puddle, but such is not the case. A number of factors replenish these fast-moving molecules until all the water in the puddle has evaporated.

These factors include

- how shallow the puddle is
- how warm the air surrounding the puddle is
- how warm the ground under the puddle is
- whether air is blowing over the puddle
- whether the puddle is in direct sunlight, and
- how low the relative humidity of the air surrounding the puddle is

As mentioned before, once water vapor molecules “nucleate” on airborne dust or bacteria, they condense out of the air and become tiny water droplets (or, with freezing, ice crystals) forming clouds. This condensation process releases the kinetic energy the water vapor molecule “stole” from the puddle, thereby warming the air around the cloud’s remaining droplets and crystals, *helping keep the cloud aloft*.

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Another factor that determines whether water evaporates from a puddle into the surrounding air is whether the air above the puddle is *already saturated* with water vapor.

For a given temperature and air pressure, **relative humidity** (RH) is

> the ratio of the current water vapor pressure to the maximum (or saturated) water vapor pressure, expressed as a percentage.

Mathematically, we write

$$RH = 100 \frac{P_{water\ vapor\ ,current}}{P_{water\ vapor\ ,saturated}} \tag{5.1}$$

For a relative humidity less than 100%, there is always an air temperature below the current air temperature where the current humidity would rise to 100%. This temperature is called the **dew point**.

If the temperature of that same air suddenly rose, the relative humidity would naturally decrease, as there would be more “elbow room” for water vapor molecules to enter the (now warmer) air.

Sometimes, in the early morning hours, when the air temperature has declined overnight, you will see dew condensed either on vegetation (like grass, or bushes) or on something else, like your car.

In the winter, relative humidity inside your home can be quite low, *even if it is snowing outside*. This is because the 100% relatively humid cold wet air outside, when let inside your home, naturally warms up and expands, and in doing so, the relative humidity in the air naturally decreases.

A **drought** can occur in a given location

> experiencing prolonged below-average precipitation.

The Sun and the Rotation of the Earth

Heating of the Earth resulting in weather events is caused almost exclusively by the Sun, which supplies a constant and renewing source of radiant energy.

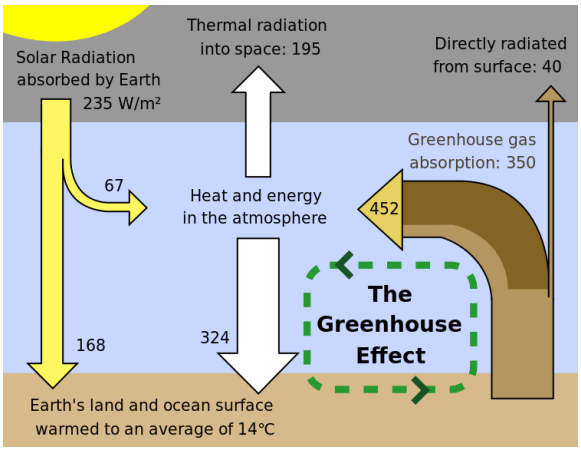


Figure 8

Some of that radiant energy warms the atmosphere directly. Other radiant energy warms the lands and seas of the Earth, which in turn warm the atmosphere.

Much of the Sun's radiant energy incident on the Earth is reflected back into space, but how much is reflected depends

> on the Earth's albedo (or reflectivity coefficient), and the Earth's greenhouse gases.

A **greenhouse gas** is a

> gas in the atmosphere that has three or more atoms in each molecule.

Prominent greenhouse gases are

> carbon dioxide, methane, water vapor, nitrous oxide and ozone.

The *right amount* of greenhouse gases is needed to keep the Earth warm enough at night so that it doesn't freeze over. Two and a half billion years ago, the Earth froze over for lack of enough greenhouse gases (methane in particular).

Too much greenhouse gas would cause the Earth's "ice cover over land" to melt, resulting in a dangerous rising of sea level (due to added water, and the expansion of water with temperature).

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Because the Earth is (roughly) a sphere, and because the Earth's axis of rotation is tilted relative to its orbit (the ecliptic), the Earth is **differentially heated**, or heated unevenly.

Since hot air rises, and cool air falls, this differential heating causes convection cells (circular loops of air) to form. Areas below rising hot air are *low pressure* areas, and areas below falling cool air are *high pressure* areas.

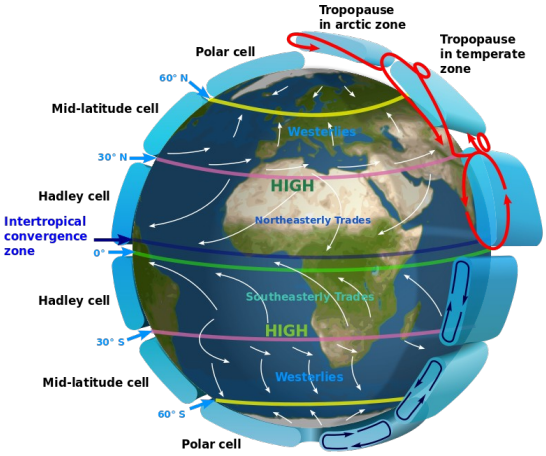


Figure 9

On Earth, for each hemisphere, three convection cells form, equator to pole, separated by the two jet streams.

The **three convection cells** are the

Hadley cell,

Ferrel or mid-latitude cell, and

polar cell

The jet streams separate these cells because of something called the Coriolis Effect, a fictitious force that, in the northern hemisphere, seems to shift northbound higher altitude winds to the east, and southbound lower altitude winds to the west, at 30° and 60° latitudes.

The circulation pattern of Earth's winds is complicated, but fortunately, the Met Office of the United Kingdom's national weather service has developed instructional videos to explain the origin of the jet streams and the underlying prevailing winds. See the **First Extended Study** for links to these videos.

At this point, it is well worth talking about temperature in greater detail.

Temperature is

An indirect measure of the average kinetic energy per molecule of the substance whose temperature you are measuring.

Temperature is *not the same thing* as heat, or internal energy. As is clear from the relatively hot temperatures seen in the stratosphere and thermosphere, a higher temperature doesn't guarantee that you will feel particularly warmer.

There are two temperature *scales* (or measurement systems) commonly used in weather measurements, the Fahrenheit (F) and the Celsius (C) or Centigrade scales.

As to how the Fahrenheit and Celsius scales came to be in their present forms, see the excellent Veritasium videos in the **Second Extended Study**.

The Fahrenheit scale is currently based on the idea that water freezes at 32° F and boils at 212° F at sea level (one "atmosphere" of atmospheric pressure, or 101,325 Pascals).

The Celsius scale, by contrast, is based on the idea that water freezes at 0° C and boils at 100° C at sea level (one "atmosphere" of atmospheric pressure, or 101,325 Pascals).

Since both scales are linear, a simple formula connecting them is

$$C = \frac{5}{9}(F - 32) \tag{5.2}$$

Volcanic Activity



Figure 10

Volcanic activity, both on land and under water, can have a significant (albeit short term) effect on Earth's climate.

When a volcano erupts, it "belches" significant amounts of fine ash and sulfur dioxide (SO_2) into the stratosphere, both of which can reduce the Earth's temperature over the short term by blocking incoming sunlight. The explosion of Mount Pinatubo in the Philippines in 1991 reduced the average global temperature the next year by 0.6° C (1° F).

Volcanoes also spew significant amounts of the greenhouse gas carbon dioxide (CO_2), which tends to raise Earth's temperature over the long term.

It is estimated that yearly output from all volcanic activity results in adding 100 to 300 million metric tons of CO_2 per year. This may sound like a lot, but it is only 1% of CO_2 added each year by humans burning fossil fuels.

Biological Activity



Video 1

It should come as no surprise to the reader that global warming, and its subsequent changes to the world's climate is a contentious subject.

Or is it?

It seems clear that human activity has greatly increased the Earth's average temperature over the last 150 years or so of data keeping, and that this increase portends significant doom for the future of the Earth, and humans thereon.

The burning of fossil fuels, deforestation, the dramatic rise in the farming of methane-producing animals for human consumption, and other human activities, are all contributing to a sharp rise in human induced greenhouse gases and the subsequent **steady, long-term increase in Earth's average temperature**. Sea levels are forecast

to rise by up to about a meter this century, and by multiple meters in the centuries beyond due to this global warming.

Watch the above video to catch a review of the available data.

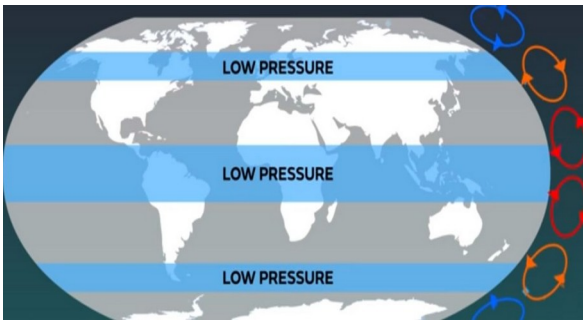
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First Extended Study: As mentioned above, the Met Office of the United Kingdom's national weather service has developed instructional videos to explain the origin of the jet streams and the underlying prevailing winds.



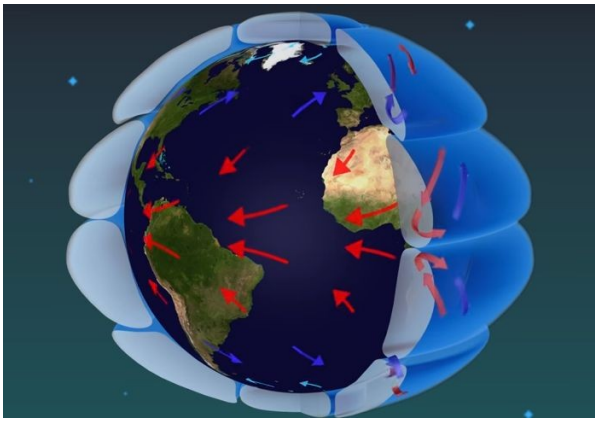
Video 2

The first video explains why there is *differential heating* of Earth's atmosphere.



Video 3

The second video explains the formation of the three convection cells in both the northern and southern hemispheres, with subsequent high and low pressure atmospheric bands.

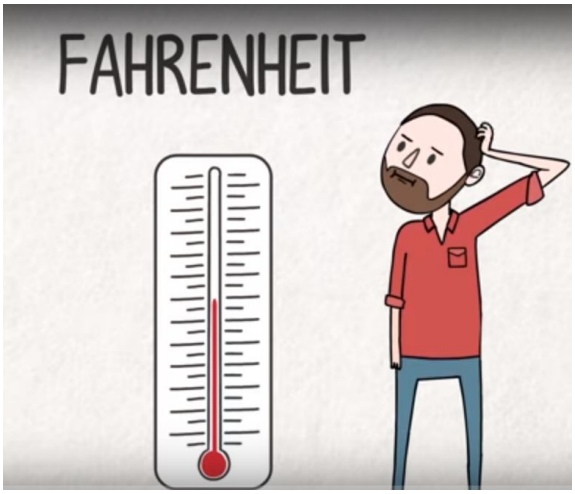


Video 4

The third video explains the formation of the jet streams and surface trade winds.

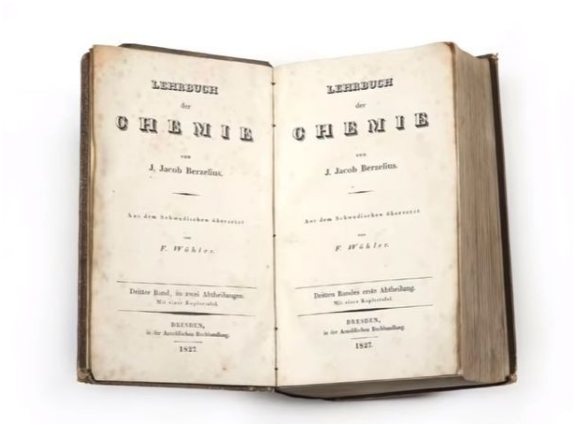
You can learn more about jet streams and their formation here.

Second Extended Study: As mentioned above, the origin of the two common temperature scales is not so obvious.



Video 5

In the first video, we learn about who Mr. Fahrenheit was, and how his scale came to be its present form.



Video 6

In the second video, we learn how Celsius came to create the temperature scale named after him.

Third Extended Study: What is space weather, and should we be concerned about it?

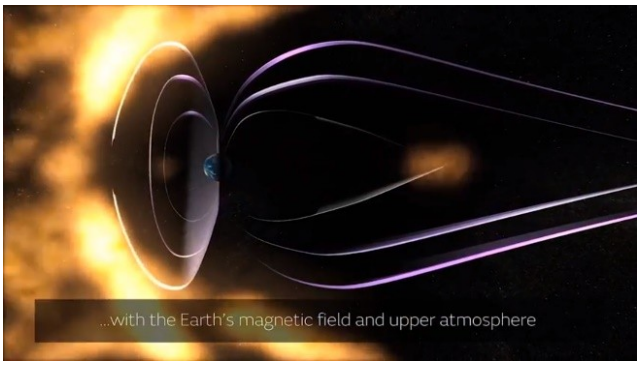


Figure 11

Space weather is the

collection of solar storm events impinging on Earth.

Besides being a source of radiative heat for Earth based weather, the Sun is a nuclear “furnace” that regularly gives off a stream of particles called the solar wind.

Sometimes the wind is associated with a spectacular explosion on the Sun’s surface, a storm of sorts, called a coronal mass ejection (CME).

In the unlikely event that a CME is heading toward Earth, significant damage can happen to satellites orbiting Earth. CMEs can also damage the Earth's power grid and disrupt high frequency radio communication used by aircraft.

The Sun is located roughly 93 million miles, or 150 million kilometers, away from Earth. To study the Sun, its solar wind, and CMEs, satellites (including SOHO) orbit a point in space called L1, about a million miles from Earth, on Earth's (always) sunny side.

Travelling at about *a million miles per hour*, the solar wind takes a little less than 4 days to get to Earth, giving engineers time to protect Earth based satellites and the power grid. While the magnetosphere can protect Earth from the regular solar wind, the advanced notice we receive from L1 satellites regarding oncoming CMEs is vital.

The above image links to yet another Met Office video on the solar wind.

HOMEWORK

1. Layers List

In case you are confused about the heights of the four layers of the Earth's atmosphere, here is a table of those heights for mid-latitudes on Earth:

Layer Name	Height, km	Height, mi	Height, ft	Pause Name
Troposphere	11	6.8	36,000	Tropopause
Stratosphere	50	31	164,000	Stratopause
Mesosphere	85	53	279,000	Mesopause
Thermosphere	600	373	1.97 million	Thermopause

Table 1

2. Ejecting Into Hell



Figure 12

The story of Lieutenant Colonel William Rankin is both fascinating and harrowing.

In the summer of 1959, Rankin was flying his F8 Crusader Jet above North Carolina at a speed of Mach 0.82 and at a height of 47,000 ft, about 2,000 ft above a threatening cumulonimbus storm cloud, when the single engine of his jet suddenly stopped working entirely.

After struggling for a while to restart the jet's engine, it was clear he was going to have to bail out over the storm cloud.

He started his journey downward into the cloud at 6 pm, without a pressurized suit, but with an external air pack. From experience, he expected to reach a safe 10,000 ft in less than 4 minutes.

On the way down, he was subjected to violent updrafts, hailstones, lightning and thunder, and rain so saturated that he thought he was going to drown in the cloud. He suffered severe frostbite and (explosive) decompression, but he eventually landed, sideways into a tree, some 40 minutes later.

After a few weeks, he fully recovered, and he subsequently wrote a book about his experience called "The Man Who Rode the Thunder."

3. Why Do Clouds Stay Up?



Video 7

Clouds are made up of tiny water droplets or crystals only a few microns in diameter, but they can weigh a great deal, making you wonder how they can possibly stay up.

The answer is buoyancy. In the presence of up-drafting air, water vapor which is less dense than the air, ascends to an altitude where it condenses into water droplets and crystals. When the droplets and crystals condense, they release the kinetic energy they "stole" when they evaporated as water vapor molecules, warming the air around them, creating an additional up-drafting pressure.

Click on the image above to see a video on this fascinating dynamic.

4. Hurricanes, Schmuricanes



Figure 13

Ever wonder what the difference is between hurricanes, cyclones, typhoons, and tornadoes? They're all highly destructive cyclonic (or circling) storms.

Tornadoes are cyclonic storms that occur over land. The other terms refer to cyclonic storms occurring over water.

Tornado strength is measured on the Enhanced Fujita Scale, from EF0 (least destructive) to EF5 (most destructive).

Cyclonic storms over water are called hurricanes, typhoons and cyclones, depending on what part of the Earth you are talking about. Functionally, they are all the same.

Hurricane is the term used in

> the North Atlantic, Northeastern Pacific, The Caribbean Sea and the Gulf of Mexico.

Typhoon is the term used in

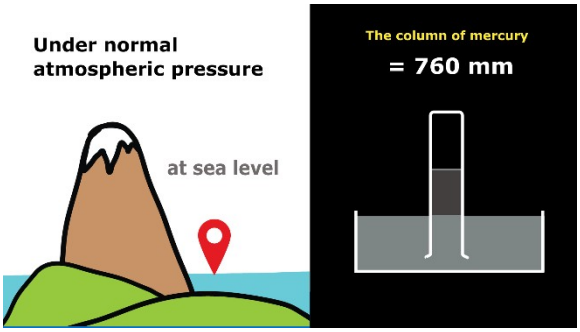
the Northwest Pacific.

Cyclone is the term used in

the Southern Hemisphere and the North Indian Ocean.

Since water has a high specific heat, and since there is a lot of it, cyclonic storms over water can pack a devastating punch. The destructive strength of such a storm is measured on the Saffir-Simpson Scale, with speed range designations similar to, but less than, the Fujita Scale.

5. How Do We Measure Atmospheric Pressure?



Video 8

Atmospheric pressure is just the

weight of the air per unit area above where you are measuring the pressure.

Atmospheric pressure can be measured using a variety of units. All the different units describe the same air pressure, but some are more convenient than others.

In this and following modules, we will use the units

Pascals (Pa), Atmospheres (Atm), millimeters mercury (mm Hg or Torr), inches mercury (in Hg), and pounds per square inch (psi).

One atmosphere (Atm) is defined as

the average air pressure at sea level.

One Atm is equal to

760 mm Hg, 760 Torr, 101,325 Pa, 29.92 in Hg, and 14.7 psi.

These units are quite general, and can be used for *any* pressure calculation, not just air pressure.

These units are *linear*, so for example, 0.5 Atm = 7.35 psi.

In general, atmospheric pressure declines with altitude, and with that decline in pressure, the boiling point of water also declines. This is because water boils when its vapor pressure equals the surrounding air pressure. For more on how pressure works in fluids, see Pascal's Principle.

For example, Chicago is at an elevation of 594 ft, relative to sea level. Its average atmospheric pressure is very close to 14.7 psi. The boiling point of water in Chicago is very close to 212° F.

Denver, by contrast, is at an elevation of 5,280 ft (one mile), relative to sea level. Its average atmospheric pressure is about 12 psi. The boiling point of water in Denver is about 203° F.

Click on the image above to see an excellent video on air pressure measurements.

6. Ancient Atmospheres



Figure 14

A little over four and a half billion years ago (4.6+ Gya), a nearby star likely exploded in a spectacular way (a supernova), seeding and compressing a cloud of gas (a nebula) that would soon enough become our solar system.

At first the cloud was haphazard in shape, but as time went on, it both flattened out and started spinning, creating a "gas and solid" disk with a bulge in the middle. That bulge would later become our Sun.

The Earth formed soon thereafter (4.543 Gya), but its earliest state was hell-like, being nearly continuously bombarded by space rocks (meteors, asteroids, comets, bolides, and planetesimals).

This bombardment period would last a good 750 million years, interrupted early on (and spectacularly) by a side-swiping collision with a protoplanet (named Theia) that would lead to the formation of our Moon (about 4.51 Gya).

The Moon coalesced out of the debris ejected by this collision. It first formed in a near Earth orbit as a ring of sorts, then came together as a sphere, our Moon.

It would take the rest of Earth's evolutionary story for the Moon to find its current distance away from Earth (385,000 km, or 239,000 miles).

As the Earth slowly cooled down, it developed a thin crust (what today we would call the lithosphere). Under the influence of gravity, the heavier elements making up Earth's insides moved to the center of the Earth, with the lighter elements moving out towards the crust (the process is complicated, but this migration is due mostly to gravity).

This migration of the heavier elements, notably iron, led to the formation of Earth's current magnetic field.

While the Earth was cooling down and forming a crust (really a series of seven major and many minor crust-like plates called tectonic plates), it was plagued with countless volcanoes spewing various gases into the atmosphere.

By contrast, any gases that *came with* the formation of the Earth, the **primordial or first atmosphere**, had long left Earth (in its inaugural phase) because these gases were

either too light for gravity's tug, or they were literally stripped away from the nascent planet by the Sun's solar wind.

[A short-lived "rock gas" atmosphere accompanied the formation of the Moon.]

The new cooled-down atmosphere, Earth's **second atmosphere**, was made up of the volcanoes' gases, which consisted primarily of carbon dioxide (CO_2), sulfur dioxide (SO_2), water vapor (H_20), and a smattering of some other gases.



Figure 15

The Earth continued to cool until the second atmosphere's temperature was below the boiling point of water (for *that* Earth's temperature and atmospheric pressure). It started to rain, and continued to do so, off and on, for several hundred million years, until, with the possible help of water-laden asteroids, comets and the like, the Earth's oceans formed, covering over 70% of our now child-like planet.

Through a process not completely understood, unicellular life called **anaerobic (or oxygen avoiding) bacteria** began sometime between 3.85 and 3.5 Gya (it may have started earlier), near cracks in the oceans' floors.

The atmosphere when life began was very dense and was mostly carbon dioxide. There was almost no oxygen or ozone in this (the second) atmosphere, and so there was no protection from the DNA destroying ultraviolet radiation raining down on the Earth from the somewhat weak but still powerful Sun. The earliest anaerobic bacteria lived by chemical synthesis (chemosynthesis), drawing energy from chemicals near cracks in the ocean floors. Later, cyanobacteria evolved a way to use sunlight to take in the then abundant carbon dioxide and produce oxygen as a waste product.

[Note: There is some debate today about **when water came to Earth**, a necessary criterion for microbial life to start and flourish. Some say that the Earth was cool enough in the Hadean eon (4.5 to 4.0 bya) to allow liquid water to form on the surface of the Earth, given its *then* supposed relatively cool air and high atmospheric pressure (230 °C, 27 Atm). Some of this speculation relates to finding life forms seemingly dating from 4.4 to 4.2 bya (see zircons).]

The oxygen produced by the cyanobacteria did not get into the atmosphere in any sizable amount for **well over a billion years**, binding instead to dissolved iron and other elements in Earth's murky oceans' waters to form solids that sank to the oceans' bottoms.



Figure 16

[These cyanobacteria used light to form oxygen from carbon dioxide (photosynthesis).

The cyanobacteria closest to the oceans' surfaces died from UV exposure, with the bacteria below it using the dead bacteria above it *as a shield* against the deadly UV radiation. This shielding ensured the continued relentless but slow production of oxygen.]

By 2.4 Gya, some of that new oxygen ascended high up into the second atmosphere, creating the first iteration of the oxygen-ozone atmospheric boundary that protects all living creatures from deadly UV radiation to this day.

This "event" or process, known as the Great Oxygenation Event, began a slow rise in oxygen lasting about two billion years, with oxygen concentration eventually peaking in the **third atmosphere** at about 30% around 300 million years ago.

[Many anaerobic organisms died off during this time, to be replaced by aerobic (or oxygen loving) bacteria.]

During this extended period, with atmospheric carbon dioxide (a greenhouse gas) on the decline, and with the newly introduced atmospheric oxygen chemically removing methane (another greenhouse gas) from the atmosphere, dramatic ice ages occurred, (see the Huronian Glaciation).

Due to continuing volcanic explosions on Earth, carbon dioxide in the atmosphere was eventually replenished, and the Earth warmed again.

Because of the continuing action of cyanobacteria, oxygen levels in the atmosphere continued to increase, and (now multicellular) life migrated from sea to land, with fungi, insects, fern-like plants, fish, and trees emerging.

With greater oxygen concentration in the atmosphere, flying insects grew to giant sizes. Finally amphibians, reptiles, dinosaurs, crocodiles, mammals and birds emerged.

While plants and cyanobacteria (some of that bacteria now hosted inside blue-green or cyan colored algae) continued to remove carbon dioxide from the air and produce oxygen.

Ultimately, all life forms (including plants at night) learned how to respire, removing oxygen from the atmosphere and producing carbon dioxide.



Figure 17

Ultimately the Earth's atmosphere settled down into its current or **fourth atmosphere**, with an oxygen level of 21%, a level that allows for fire to occur, but not spontaneously.

Carbon dioxide, once the dominant gas in Earth's atmosphere, is now a very small (but crucial) 0.04% (about 430 parts per million, or ppm).

Today, many atmospheric scientists are concerned that increases in carbon dioxide in Earth's atmosphere -- increases due primarily to the burning of fossil fuels -- will lead to an overheated Earth, rising sea levels, disruptions in the jet streams and ocean currents, and catastrophic weather patterns.

7. Where Did All the Argon Come From?



Figure 18

The third most abundant gas in our (dry) atmosphere is Argon, specifically the isotope Argon-40. It comes from the radioactive decay of Potassium-40. Since Potassium-40 has a half life of 1.248 billion years, and Argon-40, is an inert and stable gas, it is reasonable to assume that the Argon in Earth's atmosphere has accumulated over the evolution of the Earth's atmosphere (4.5 billion years).

8. How Adverse Weather Affects Flight

Modern commercial jet airplanes are quite robust, but safety in the presence of adverse weather conditions is always a consideration. Less than stellar flying weather includes rain, high cross-winds at time of landing, microbursts, clear air turbulence, icing of a plane's wings when flying or just before takeoff, clear runways during a snowfall, and diminished visibility due to fog.

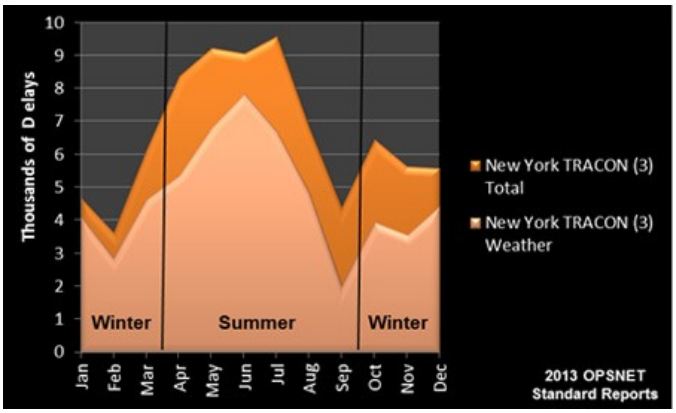


Figure 19

Counterintuitively, the Federal Aviation Administration (FAA) reports more flight delays in summer months as opposed to winter months, with 69% of all flight delays in summer due to adverse weather conditions, primarily violent rain storms, low cloud ceilings with impaired visibility, and high winds blowing across runways.

High air temperature can on rare occasion affect whether a large plane *has enough lift* to safely take off.

Pilots always try to fly around storms rather than above them, but modern commercial jet aircraft are well designed to handle storm conditions.

One exception is ice accumulating on a plane's wings, either before takeoff, or during flight, with many modern planes equipped to pneumatically break up any ice that accumulates on a plane's wings during flight.



Video 9

Another exception are microbursts, strong downward gusts of wind of weak tornadic strength that can disrupt a plane's flight in dangerous ways, especially at landing. Here is a video of Delta Flight 191, a Lockheed L1011 Tristar, attempting a landing at Dallas Fort Worth International Airport on August 2, 1985.

Additionally, pilots sometimes have to contend with clear air turbulence, a violent wind buffeting event that occurs when planes cross fast and slow moving streams of air.

9. How Strong Could Atmospheric Pressure Actually Be?



Video 10

Earlier, we learned that atmospheric pressure in the US-English system of units is 14.7 psi. That sounds like a lot!

A regular sheet of letter-sized paper is 93.5 square inches, so the *force* the atmosphere presses down on the paper with is 1,374 lbs. How can you even lift a single sheet of paper off a table?

The answer is that the air pressure above and below the paper are the same, so the *pressure difference* between the top and bottom of a letter sized sheet of paper is essentially zero.

The above video cleverly reveals just how strong air pressure is, when there is a difference between one side of something, like an egg, and its other side.

10. How Ozone and Oxygen Protect Us

Shortwave UV light photons (100-240nm) have enough energy to knock electrons in oxygen molecules up to higher orbital levels, creating excited oxygen molecules, which then decompose to pairs of so-called free radical oxygen atoms. Each “free radical” oxygen atom then combines with an unexcited oxygen molecule, creating a net of two ozone (O_3) molecules.

The bottom line here is that electrons are not "knocked out" of the excited oxygen molecules: they are just moved to higher energy orbits of the oxygen molecules.

Since orbital energy transitions are quantized (numerically specific), it takes a very specific energy (photon frequency) to make one of these orbit jumping transitions happen.

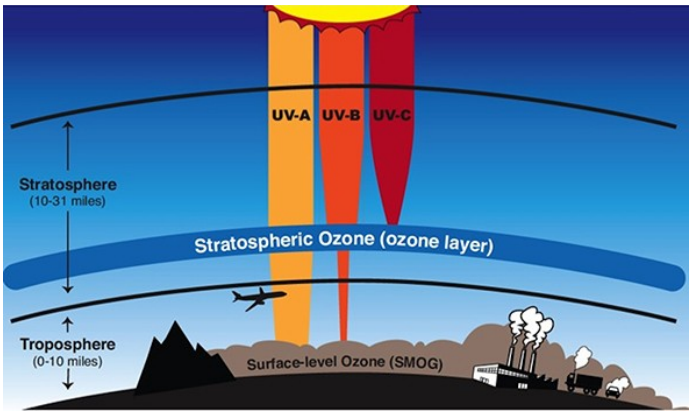


Figure 20

Now there is a second reaction to consider as well. Longwave UV light photons (240nm - 350nm) have enough energy to excite an ozone molecule, which then breaks apart into a free radical oxygen atom and an oxygen molecule.

In summary, Earth's ground inhabitants (us) are protected from shortwave UV by oxygen, and are protected from longwave UV by ozone. Neither alone would protect us from all UV effectively.

Now the rest of the story is that not all oxygen molecules bouncing around in the air are in the same energy state due to atom-atom vibrations. So the UV absorption occurs as a "band" rather than as a "fine line" like in the absorption spectra of Earth's atmosphere.

Notes:

UVA = 315 to 400 nm UVB = 280 to 315 nm UVC = 100 to 280 nm

UVA generally gets to Earth ground undisturbed by the upper atmosphere. UVA causes little sunburn, but it does age the skin and contributes to skin DNA damage. The little UVB that gets past the ozone layer can contribute to sunburn, and it can directly damage skin DNA thus causing skin cancer. While UVB is partially blocked by clouds, depending on cloud type, it is **still possible to get sunburned on a cloudy day**.

11. Clockwise High, Counterclockwise Low

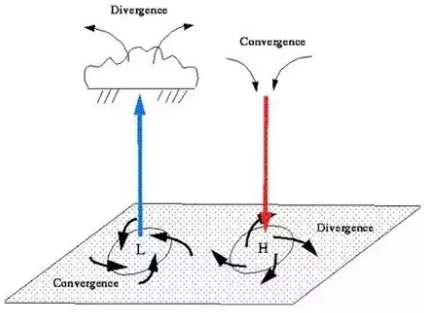


Figure 21

What causes high and low pressure spots in Earth's weather system?

Why do high pressures rotate clockwise, but low pressures rotate counterclockwise?

Glad you asked!

Hot air rises and cold air falls, in accordance with Archimedes' Principle of Buoyancy.

When hot air rises, it leaves behind a partial vacuum, or low pressure area.

When cool air sinks, it creates a high pressure underneath it.

In short:

"Low pressure results from rising hot air, high pressure results from falling cold air."

When hot air rises, the water vapor molecules in it naturally condense at cooler higher altitudes (as that air parcel reaches its dew point temperature). Clouds and precipitation thus ensue.

This is why low pressure is associated with rain, snow, etc.

When cold air falls, it suppresses upward water vapor movement (by evaporation), so little moisture ultimately gets into the air, and little precipitation occurs.

This is why high pressure is associated with clear skies and no rain, snow, etc.

High pressures *rotate clockwise* (anticyclonic rotation), and low pressures rotate *counterclockwise* (cyclonic rotation). Both rotations are due to the **Coriolis Effect**.

In the Northern Hemisphere, the Coriolis Effect deflects air to the right of which way the air is moving. For **high pressure** columns of air, the air is moving downward and then outward. Deflection to the right of ground-level outward moving air is thus **clockwise**, when viewed from above (like you'd see on a weather map).

For **low pressure** columns of air, the air is moving upward and inward. Deflection to the right of ground-level inward moving air is thus **counterclockwise**, when viewed from above.

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